Does Domestic Violence Law Shift Gender Attitudes and Female Labor Force Participation in Kyrgyzstan?*

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Work in progress. Comments welcome.

Abstract

Reducing gender gaps in labor market outcomes is a central policy priority worldwide. This paper examines whether legal reforms aimed at protecting women can translate into substantive economic empowerment in contexts characterized by strong patriarchal norms. Using the Domestic Violence Law in Kyrgyzstan, I study its effects on gender attitudes, intra-household decision-making power, and female labor force participation (FLFP). Using an intensity-based triple-differences design that exploits regional variation in pre-reform domestic violence prevalence, I find that the law led to significant improvements in attitudes toward women's work and increased women's bargaining power within households. These effects are particularly pronounced in initially conservative regions with higher exposure to domestic violence prior to the reform. However, I find no increase in FLFP across regions. This divergence between changing norms and unchanged economic behavior highlights a key limitation of legal empowerment: while anti-violence legislation can reshape perceptions and private decision-making, structural and social constraints may continue to limit women's observable labor market participation. The findings contribute to broader debates on discrimination, gender norms, and the conditions under which legal reforms translate into meaningful substantive economic outcomes.

Keywords: Female Labor Force Participation; Domestic Violence; Perception

JEL: D1, J16, K38

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1 Introduction

Reducing gender gaps in labor market outcomes is a central policy priority worldwide. Despite substantial progress in education and legal reforms, large gender disparities in female labor force participation (FLFP) persist across countries. In Kyrgyzstan, for instance, female educational attainment at the lower secondary level is comparable to that of men, with completion rates of 94.4% for girls and 95.7% for boys. Yet, female participation in the labor market lags far behind: FLFP is only 53.5%, compared to 79% for men (World Bank, 2023). This pattern indicates that even in the presence of relatively equal educational attainment, formal institutions may not be sufficient to eliminate gender gaps in labor market participation.

What explains this persistent disparity? While firm-side interventions such as anti discrimination policies are important for expanding formal employment opportunities, they may be insufficient when household constraints remain binding. Beyond labor market frictions, multiple factors including labor market frictions, childcare availability, and household resources affect women's labor supply, recent research highlights social norms as a key mechanism operating within households.[1] Traditional gender-role norms are associated with lower FLFP and reduced working hours (Bertrand et al., 2015; Alesina et al., 2013; Lundberg, 2024). In many contexts, these restrictive norms affect not only attitudes but also daily practices that restrict women's autonomy. In Kyrgyzstan, domestic violence, including long-standing practices such as bride kidnapping (ala kachuu), remains prevalent and is associated with severe psychological and socioeconomic consequences that limit women's autonomy and economic participation(Arabsheibani et al., 2021; JICA and EY, 2022).

Importantly, domestic violence directly affects intra-household bargaining by weakening women's outside options and decision-making power(Heath, 2014; Anderberg et al., 2016). As a result, gender norms may operate through violence and control within households to suppress women's labor supply. Building on this, Lundberg (2024) emphasizes that when preferences and beliefs are malleable, legal institutions can influence behavior not only through punishment but also by reshaping underlying norms.

This raises the question of whether modern legal reforms can weaken entrenched norms and promote women's participation in the labor market. In line with this view, Kyrgyzstan introduced a series of gender-related legal reforms, including the criminalization of child religious marriages in 2016, the adoption of the Domestic Violence Law in 2017, and the classification of domestic violence as a criminal offense in 2019 (Baker McKenzie, nd; International Alert, 2020). This study investigates whether Kyrgyzstan's domestic violence law reforms, introduced in 2017 and strengthened in 2019, acted as a catalyst for changing gender attitudes and increasing women's labor market participation.

To estimate causal effects, I apply a triple-differences (DDD) framework, exploiting regional variation in baseline DV prevalence as a proxy for treatment intensity. Since the DV Law was

[1]Historical origins may also matter. As Alesina et al. (2013) point out, in societies where plough agriculture historically dominated, farming required physical strength and thus favored male labor. Women were pushed into domestic tasks and child-rearing, reinforcing a gendered division of labor that eventually solidified into social norms. This suggests that some countries undergoing economic transition may still carry norms rooted in historical production technologies. At the same time, it is difficult to disentangle whether current constraints stem from such historical legacies or from contemporary institutional and economic factors, making the analysis of FLFP particularly challenging. Therefore, rather than disentangling historical versus contemporary institutional constraints, this study focuses on social norms as a key mechanism influencing women's labor force participation.

implemented uniformly nationwide, oblasts with higher baseline DV exposure are plausibly more affected. This baseline heterogeneity largely reflects stable institutional and reporting capacity rather than individual choices, providing a credible source of exogenous variation for identification.

I used a longitudinal panel data from the Life in Kyrgyzstan (LiK) Study, which offers unique advantages for this analysis. Its panel structure allows controlling for individual fixed effects, reducing concerns about unobserved heterogeneity. Moreover, the survey includes rich information on both labor market outcomes and gender-related attitudes, enabling me to link legal reforms not only to behavioral changes but also to shifts in norms. Crucially, the data cover both pre- and post-reform periods, making it possible to exploit regional variation in baseline DV prevalence

The results reveal a clear divergence between attitudes and behaviors. The DV Law significantly increased women's support for female employment and strengthened their decision-making power within households, particularly among women in oblasts with higher pre-reform DV prevalence. However, these normative and intra-household improvements did not translate into measurable changes in women's actual labor market participation or risk-taking behavior. Taken together, the evidence suggests that the law primarily operated through attitudinal and bargaining channels, while structural constraints continued to limit behavioral responses.

This study contributes first to the literature on social norms and female labor supply. A large body of research shows that cultural attitudes shape women's participation in labor markets: for example, Fernández and Fogli (2009) show that among U.S.-born women of immigrant parents, labor supply and fertility outcomes are significantly shaped by the cultural environment of their parents' country of origin; Alesina et al. (2013) show that the historical adoption of agriculture, reinforced a gendered division of labor that solidified into persistent social norms about women's roles; and Bertrand et al. (2015) demonstrate labor supply responses to breadwinner norms in the U.S.

More recently, Bursztyn et al. (2020) illustrate how correcting misperceived norms can meaningfully increase women's job applications in Saudi Arabia, and Cavapozzi et al. (2021) show that mothers embedded in peer networks with more progressive norms are significantly more likely to participate in the labor force. Finally, Beaman et al. (2009) show that repeated exposure to female leaders through political quotas in India improved perceptions of women's leadership effectiveness and weakened some gender-role stereotypes, although deeper preferences for male leaders remained more resistant to change. Similarly, La Ferrara et al. (2012) show that exposure to Brazilian telenovelas, which portrayed small families and empowered female characters, led to lower fertility, suggesting that media can shift norms and behavior even without formal mandates. By documenting how DV law reform reshaped attitudes toward female employment, this study extends the literature by highlighting how legal institutions can act as normative signals in transitional societies.These findings support and complement existing work on cultural determinants by demonstrating that normative change can also be observed in response to institutional shocks.

In many cases, cultural expectations dictate that men should be the primary breadwinners, while women's roles are confined to the domestic sphere. These beliefs can not only constrain women's employment decisions but also legitimize male control, including through violence (Jayachandran, 2021).

Second, this paper is also related to the literature on the causal effects of the legal reform of DV laws on gender attitudes and women's labor market outcomes. Gedikli et al. (2023)

suggest IPV may increase FLFP in Turkey; Lenze and Klasen (2017) find no clear causal effect in Jordan; and García-Ramos (2021) reports increased IPV after divorce law reforms in Mexico. By contrast, Gu et al. (2022) find China's Anti-DV Law improved women's well-being and workplace outcomes. These findings underscore that DV and women's economic agency are linked through context-specific mechanisms. By focusing on Kyrgyzstan's reform, this study adds new evidence from Central Asia—an under-researched region where institutional change and entrenched patriarchy intersect.

More broadly, this paper underscores both the power and the limitations of legal institutions in catalyzing social change. By tracing impacts not only on outcomes but also on attitudes, it highlights how law can function not merely as deterrence but as a signal that reshapes normative environments. These insights are essential for policymakers seeking to promote women's economic inclusion in societies where cultural and institutional barriers remain deeply intertwined.

The remainder of this paper is structured as follows. Section 2 provides background information on Kyrgyzstan, focusing on the institutional context surrounding FLFP and domestic violence. Section 3 presents the conceptual framework. Section 4 describes the data and reports descriptive statistics. Section 5 presents the empirical findings. Section 6 discusses the results. Section 7 concludes and outlines the policy implications.

2 Background

2.1 Background of Kyrgyzstan

Kyrgyzstan is a transitional society in which rapid institutional change has occurred alongside the persistence of deeply rooted kinship-based social structures. Historically, Kyrgyz society evolved from nomadic, patrilineal family systems into Soviet collectivization under *kolkhoz* and *sovkhoz* structures, and later into a market economy. Rather than being dismantled, extended patrilineal kinship networks (*uruk*) were reorganized under socialism and continue to structure social and economic interactions today. These networks function asymmetrically by gender: while they provide men with informal insurance and social security, they often constrain women by enforcing expectations about appropriate behavior within marriage and the household(Zhussipbek and Nagayeva, 2021). Consequently, norms governing women's roles are reproduced within families even when formal institutions and market incentives change.

At the same time, Central Asian countries adopted different paths of economic and political transition. While for example Uzbekistan pursued a gradual, state-led development strategy often referred to as the "Uzbek model," Kyrgyzstan implemented rapid market-oriented reforms based on so-called "shock therapy". These reforms were accompanied by prolonged political instability, including repeated episodes of social unrest and regime change until around 2010.

Despite these institutional and political challenges, Kyrgyzstan was relatively early among post-Soviet Central Asian countries in introducing specialized legislation, victim protection, and criminalized mechanisms. Exposure to globalization, legal reform, and international gender-equality discourse has made it increasingly easy for people to adopt egalitarian gender norms.

For these reasons, the combination of persistent traditional norms, rapid institutional tran-

sition, and early legal reform makes Kyrgyzstan a particularly informative setting for examining the interaction between law, social norms, and women's economic behavior, as well as the economic effects of legal protection for women. At the same time, it shares key characteristics with many post-socialist countries, making it a broadly relevant case for understanding the conditions under which legal reforms translate into economic outcomes.

2.2 DV law in 2017 and 2019

In Kyrgyzstan, formal legal regulation of domestic violence began with the adoption of the first Domestic Violence Law in 2003. While this early legislation introduced formal protection mechanisms for the first time, domestic violence continued to be treated primarily as a private family matter in police and judicial practice. As a result, victim protection and perpetrator sanctions remained largely ineffective in practice.

A major institutional shift occurred with the adoption of the Law of the Kyrgyz Republic On Prevention and Protection Against Domestic Violence in April 2017 (Law No. 63)hereafter referred to as the DV Law. The 2017 law substantially expanded the institutional framework for victim protection by introducing emergency and long-term protection orders, mandatory reporting obligations, and formal mechanisms for inter-agency coordination among law enforcement, social services, and local authorities (UN Women, 2017; Baker McKenzie, nd). In contrast to the earlier legislation, the 2017 reform clarified institutional responsibilities and established standardized procedures for protection and reporting, in line with international human rights obligations.

While the 2017 law strengthened procedural and protective mechanisms, criminal sanctions against perpetrators remained limited. This changed with the 2019 reform, which formally classified domestic violence as a criminal offense. By introducing stronger punitive measures, the 2019 amendment increased the legal costs faced by perpetrators and enhanced the credibility of the institutional framework established in 2017. Together, the two reforms marked a transition from a largely administrative and mediation-based approach to a more formalized and enforcement-oriented legal regime.

Given the timing of the LiK survey waves, which span both the pre-reform and post-reform periods, this study treats the 2017 and 2019 reforms as a combined institutional intervention. This approach reflects the fact that the two reforms jointly altered both formal protection mechanisms and enforcement incentives, allowing the analysis to capture the overall impact of strengthened legal protection on gender attitudes and women's economic behavior.

3 Conceptual Framework: Law, Normative Pressure, and Women's Labor Decisions

This study draws on household bargaining theory and the concept of normative pressure to understand how legal reforms, particularly domestic violence (DV) legislation, may influence women's labor force participation (FLFP). Rather than treating legal change as merely a punitive tool, the analysis considers its potential to reshape social expectations and intra-household dynamics. In the standard collective household model, each member's utility is influenced by both their individual preferences and their bargaining power, which in turn depends on outside options or threat points (Lundberg and Pollak, 1993). When external institutional support,

such as access to legal divorce, police protection, or social services, enhances a woman's fallback position, her bargaining power within the household may increase. This shift can enable her to negotiate more effectively on matters such as labor market participation. In the context of Kyrgyzstan, the DV Law may contribute to this shift by enhancing women's fallback position and bargaining power, assuming effective implementation and access to institutional support. The core hypothesis of this study is that the DV Law may enhance women's ability to make autonomous labor market decisions by improving their bargaining position and perceived legitimacy, particularly through legal protection and institutional support. Importantly, even where enforcement remains imperfect, the law may also function as a source of normative pressure. By publicly declaring domestic violence as socially and legally unacceptable, it may signal a broader societal shift. According to Lundberg (2024), laws can serve as signals that define what is morally acceptable, thereby shaping both behaviors and values. In this sense, the DV Law may have encouraged communities to reevaluate previously tolerated behaviors and offered women a stronger sense of legitimacy in asserting their rights.

The normative shift may affect women's labor force participation through two main pathways. First, by reducing the risk of violence or coercion within households, the law may enhance women's sense of safety and autonomy, making employment a more feasible option. Second, by shifting social perceptions among both women and men regarding the legitimacy of women's roles in the workforce, it may contribute to reducing stigma and improving women's confidence and economic agency over time. While the law may operate differently across genders, functioning as a threat for potential perpetrators and a safeguard for potential victims, the present study focuses on its implications for women's labor market behavior.

The empirical analysis does not estimate a structural model, but instead applies a reduced-form difference-in-difference-in-differences (DDD) approach to estimate the average effect of the DV Law on FLFP. The theoretical framework thus motivates the empirical hypothesis and guides the interpretation of results by framing domestic violence as both a manifestation and a determinant of bargaining power.

Crucially, the bargaining and signaling mechanisms outlined above are not co-equal; they imply an ordering of expected responses. They act first on the margins closest to the household—a woman's autonomy over intra-household decisions and the gender-role attitudes that she and those around her hold. Labor force participation is a downstream outcome of these proximate shifts, and the pass-through is far from automatic: occupational segregation, widespread informal employment, and employer bias—conditions women do not control—attenuate the translation of greater bargaining power into observed work. The framework therefore predicts a specific signature in the data: the reform's footprint should be clearest in decision-making and attitudes, and weak or lagged in aggregate labor supply. A split of this kind—movement upstream alongside a muted labor response—is what the theory predicts, not evidence against it.

4 Data

4.1 Data Source

This study uses panel data from the Life in Kyrgyzstan (LiK) survey, a nationally representative longitudinal dataset that tracks individuals aged 18 and above. I use three survey waves

from 2013, 2016, and 2019. The 2016 and 2019 waves bracket a period of major legal change, during which Kyrgyzstan implemented the combined DV legal reforms.The final balanced panel consists of 6,874 individual-year observations between 2016 and 2019.

To ensure consistent treatment assignment and reduce measurement error in exposure to the DV Law, I exclude individuals who changed their oblast of residence between 2016 and 2019. This restriction ensures that each individual is consistently assigned to a single treatment unit, minimizing potential misclassifications of treatment intensity. Only 29 individuals were dropped due to this restriction, indicating that the impact on the overall sample composition is minimal.

4.2 Key Variables

Five types of outcome variables are used in this analysis. First, the behavioral outcome is a binary employment indicator equal to one if the respondent reports any work activity in the past seven days. Following the Life in Kyrgyzstan employment module, this includes work for an employer outside the household, work on a household-owned or -rented farm or business, subsistence farming, fishing, hunting, or gathering, and unpaid family work. It is therefore a broad measure of economic activity rather than of paid employment alone: it does not distinguish wage work from self-employment, subsistence agriculture, or unpaid family labor. This variable is defined for both male and female respondents. To capture gender heterogeneity, I include an interaction term which is DDD term with the female dummy. This allows me to assess differences in responses between men and women, with particular attention to female labor force participation (FLFP).

Second and Third, perception-based outcomes on gender roles are derived from two key attitudinal statements:
"A husband's career should be more important to the wife than her own"
"A woman should not work outside her home due to religious considerations"
Each is measured on a 4-point Likert scale (1 = strongly disagree, 4 = strongly agree), with higher values indicating stronger endorsement of traditional gender norms. Gender norm dummies capturing strong agreement (score = 4) with either "husband's career should come first" or "women should not work for religious reasons" in 2016. These covariates help account for baseline heterogeneity in attitudes and demographics that may influence labor and perception outcomes. Figure 1 displays the proportion of respondents who agreed or disagreed with the statement: "A husband's career should be more important to the wife than her own," by year and gender. The chart shows a decrease in strong agreement from 2013 to 2016, followed by a rebound in 2019. Notably, male respondents were consistently more likely to strongly agree with the statement compared to female respondents, confirming the t-test results reported above. These patterns are also illustrated visually in Figure 2, which shows the frequency distribution of responses over time by gender.

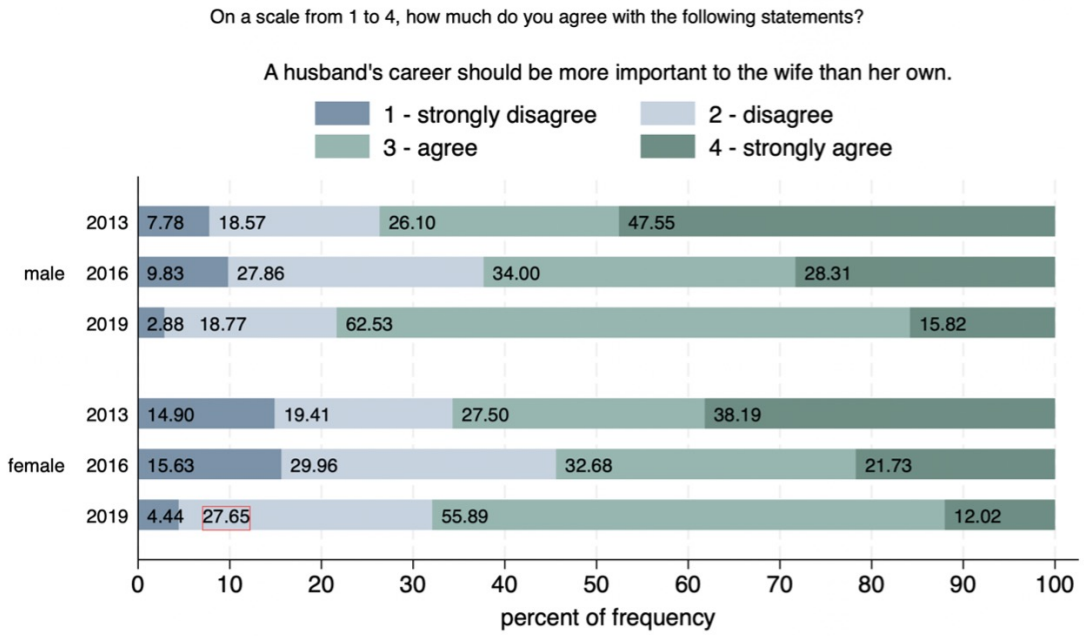


Figure 1: The proportion of people who believe the husband's career is more important than the wife's by year and gender.

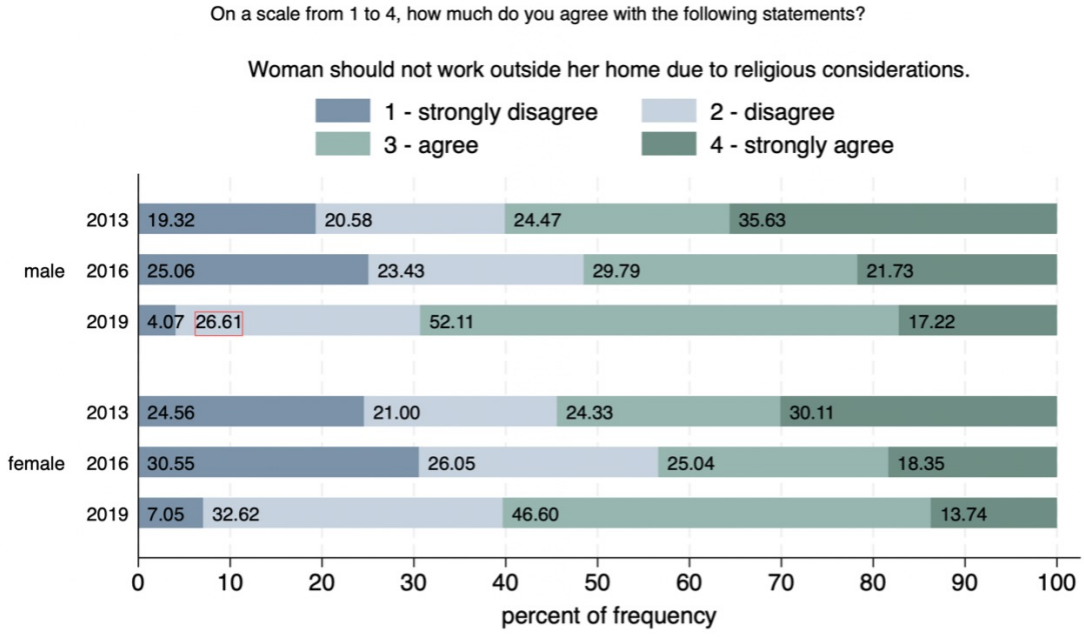


Figure 2: The proportion of people who believe woman should not work outside by year and gender.

Fourth, the risk attitude measure is based on the survey question: "How do you see yourself? Are you generally a person who is fully willing to take risks, or do you avoid taking risks?" Responses range from 0 ("completely unwilling to take risks") to 10 ("completely willing to take risks"). This variable serves as a proxy for individual risk aversion.

$$\text{Risk std}_{i,t} = \frac{\text{Risk}_{i,t} - \overline{\text{Risk}}_{2016}}{SD(\text{Risk}_{2016})} \tag{1}$$

This risk attitude variable is standardized to have a mean of 0 and a standard deviation of 1 in 2016. This scaling allows coefficients to be interpreted as changes in standard deviations relative to the baseline distribution. Using this measure, I examine whether the DV Law altered general risk attitude. The theoretical interpretation is twofold. First, an increasing risk attitude may indicate that women perceive greater institutional protection after the law was enacted. In this case, a stronger sense of safety lowers the psychological barriers to working outside the home or participating in household decision-making. Second, it is also possible that risk attitude decreases.The DV Law may have raised awareness that domestic violence is a serious issue, thereby making risks more visible. In this scenario, a higher score reflects heightened awareness of potential risks rather than greater willingness to take action.

Lastly, I constructed a variable on women's decision-making power within the household as a measure of empowerment. Due to data limitations, it is not possible to precisely identify which female household member is the wife within each household. Therefore, in this study, women's empowerment is measured using an indicator of female decision-making power, coded as 1 if any female household member is reported to participate in decision-making in a given domain. Decisions attributed to "parents" are excluded from this measure, as the gender of the decision-maker cannot be identified. The final index is constructed as the average share of decisions (out of 17) in which at least one female household member participated, ranging from 0 to 1. This variable should be interpreted, in line with the theoretical framework, as a proxy for women's bargaining power within the household.

Treatment intensity is defined at the oblast level using pre-law DV prevalence in 2016. Specifically, I construct an incidence rate:

$$DV\ rate_{o,2016} = \frac{\text{Number of reported female DV victims in oblast in 2016}_o}{\text{Estimated female population in oblast in 2016}_o} \times 100 \tag{2}$$

To facilitate interpretation of interaction terms, I made the DV incidence rate mean-centered around its 2016 oblast-level average, so that the post coefficient can be interpreted at the average level of DV prevalence:

$$DV\ rate^{c}_{o,2016} = DV\ rate_{o,2016} - \overline{DV\ rate_{2016}} \tag{3}$$

where $\overline{DV\ rate_{2016}}$ denotes the sample mean of oblast-level DV rates in 2016. With this transformation, the coefficient on the post-year dummy can be interpreted as the effect for an oblast with an average prevalence of DV. The DV rate is expressed in percent of the female population. For ease of interpretation, I rescale the centered 2016 DV rate as $DV\ rate^{s}_{2016,c} = DV\ rate_{2016,c}/0.10$, so that a one-unit increase corresponds to a 0.1-percentage-point increase in the DV rate, roughly one standard deviation across oblasts (0.099 percentage points). I use this rescaled, centered variable, referred to as "DV rate", in all regressions in the paper.

Total population data were obtained from open data portal; National Statistical Committee of the Kyrgyz Republic (2025). Female population estimates were manually constructed by dividing the total oblast population by two for each year, based on the approximately equal gender distribution reported in National Statistical Committee of the Kyrgyz Republic (2021). Data on reported DV cases were obtained from public reports of the National Statistical Committee of the Kyrgyz Republic (2024).

These administrative statistics such as DV victim case reports are likely affected by reporting bias, especially in rural areas. Consequently, higher reported incidence in urban oblasts such as Bishkek likely reflects greater access to reporting channels rather than higher underlying prevalence. Because the measure is based on reported cases, it may reflect the ease of reporting rather than the true underlying prevalence of domestic violence. In urban oblasts such as Bishkek, where access to reporting channels is greater, the incidence rate is likely to appear higher. By contrast, in rural areas the number of reported cases may be severely underestimated due to stigma, limited access to services, and institutional distrust. Therefore, the DV rate should be interpreted not as the actual prevalence of domestic violence but rather as a proxy for the intensity of reported cases.

Additionally, figure 3 shows the annual trend of DV among women incidence rates by oblast. The vertical axis represents the DV incidence rate (%) among women in each oblast, while the horizontal axis indicates the years from 2012 to 2022. The line types and colors correspond to different regions, including oblasts and major cities. The black dashed line marks the year 2017, indicating the implementation of the DV law as a policy intervention.

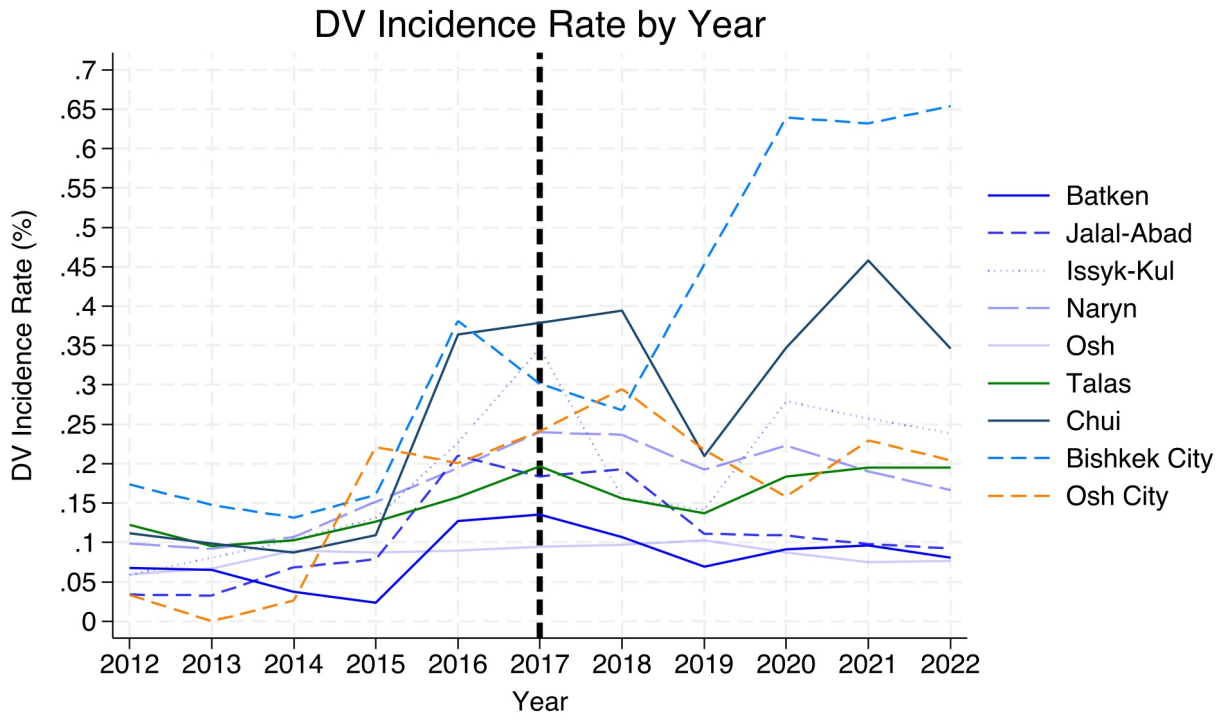


Figure 3: The annual trend of DV incidence ratesa among women by oblast.

I include a set of individual-level covariates commonly used in gender and labor studies:

Marital status (1 if married in 2016, 0 otherwise) and Age and Age^2. Gender norm dummies capturing strong agreement (score = 4) with either "husband's career should come first" or "women should not work for religious reasons" in 2016. These covariates help account for baseline heterogeneity in attitudes and demographics that may influence labor and perception outcomes.

4.3 Descriptive Statistics

Table 1 presents the summary statistics for the variables. Variables include demographics, labor force participation, perception items and DV indicators.

Table 1: Summary statistics of all samples in main analysis

	Mean	Std. dev.	Min	Max
Outcomes				
Work (past 7 days)	0.506	0.500	0	1
No WWOH	2.564	0.962	1	4
Husband career importance	2.756	0.860	1	4
Risk attitude (std., 2016=0)	-0.219	1.124	-1.890	2.099
Female decision-making index	0.574	0.365	0	1
Treatment and demographics				
DV rate (2016, %)	0.206	0.099	.0895	0.381
DV rate mean-centered (2016, %)	0.000	0.099	-0.116	0.175
Female	0.562	0.496	0	1
Post year(=2019)	0.500	0.500	0	1
Average sample size per oblast	764	539.3	266	1,674
Baseline covariates (2016)				
Number of children (<18)	2.134	1.653	0	9
Age	43.031	15.384	18	92
Log food consumption	11.553	0.446	9.989	13.703
Log non-food consumption	13.355	0.806	10.086	16.203
Married	0.780	0.414	0	1
Restrictions(=1 if strongly agree in 2016)				
No WWOH (2016)	0.196	0.397	0	1
Husband career importance (2016)	0.245	0.430	0	1
Observations	6,874 (6,778 for consumption variables)			

Table 1 shows that approximately 50.6% of respondents reported working in the past seven days, reflecting relatively low levels of labor force participation in the sample. The perception-based outcomes indicate that, on average, respondents expressed moderate agreement with traditional gender norms: the mean score for "women should not work outside the home" is 2.56 (on a 1–4 scale), while "husband's career importance" has a mean of 2.76.

The standardized risk attitude variable has a mean of –0.219, confirming that values were rescaled relative to the 2016 baseline (mean = 0, SD = 1). The female decision-making index averages 0.574, suggesting that women participate in roughly 57% of household decision items. The treatment variable, oblast-level DV prevalence in 2016, averages 0.206 percent of the female population (about 206 reported female victims per 100,000 women), but exhibits substantial variation across regions (SD = 0.099 percentage points, ranging from 0.09 to 0.38 percent). About 56% of the sample are female, and roughly half of the observations are from 2019 (post-reform period). Baseline covariates highlight that the average respondent in 2016 was 43 years old, with around two children, and 78% were married. Household consumption is measured using self-reported expenditures over the past 12 months. Specifically, respondents report average household spending on food and non-food items during the previous year, which are log-transformed and used as proxies for economic status. Finally, the restrictions section indicates that, in 2016, 19.6% of respondents strongly agreed with the statement that women should not work outside the home, and 24.5% strongly agreed that the husband's career should take priority. These subgroups provide useful variation for analyzing heterogeneity in normative attitudes and for testing whether the reform affected those with more conservative baseline views.

Table 2: Comparison of means between 2016 and 2019 (two-sample t-tests).

Variable	Mean 2016	Mean 2019	Diff (2016–2019)	p-value
Work (work7)	0.483	0.528	-0.045***	0.0002
No WWOH reliance	2.396	2.732	-0.336***	0.0000
Husband career importance	2.689	2.823	-0.134***	0.0000
Risk attitude (std., 2016=0)	0.001	-0.440	0.441***	0.0000
Female decision-making index	0.572	0.577	-0.005	0.5586
Married (=1)	0.780	0.726	0.054**	0.0127
Female (=1)	0.562	0.562	0.000	1.0000

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2 shows that several key outcome variables experienced significant changes between 2016 and 2019. First, labor force participation (Work) rose from 48.3% in 2016 to 52.8% in 2019, a statistically significant increase, indicating that overall participation expanded over the period. Average agreement with traditional gender-role statements also rose. Endorsement of the belief that women should not work outside the home increased from 2.396 to 2.732, and agreement that a husband's career should be prioritized over the wife's rose from 2.689 to 2.823. Because higher scores denote stronger endorsement of traditional norms, these unconditional movements indicate that stated norms became, if anything, more traditional on average—consistent with the post-2016 rebound in Figure 1. The standardized risk-attitude score declined from 0.001 to −0.440, about 0.44 standard deviations below the 2016 baseline; because higher values denote greater willingness to take risk, this indicates that respondents became, on average, less willing to take risk (more risk-averse) over the period. These are unconditional, pooled comparisons and are descriptive only: they combine men and women across all regions and confound the reform with every other change over the period—including, for gender norms, an apparent national rebound toward more traditional attitudes. The

causal question—whether outcomes evolved *differentially* for women in higher-DV-prevalence regions—is addressed by the DDD estimates below, which net out these common trends and, for gender attitudes, recover a significant shift in the opposite (less traditional) direction for treated women.

5 Empirical strategy

To examine whether the combined DV legal reforms affected female labor force participation (FLFP) and perceptions regarding FLFP, I implement a two-period, two-way fixed effects intensity Difference-in-Difference-in-Differences (DDD) approach with wild-cluster bootstrapping, using panel data from 2016 and 2019 for the main analysis. In addition, using data from 2013, 2016, and 2019, I conduct a pre-trend analysis for the 2013–2016 period to validate the parallel trends assumption prior to the reform. The specification follows the approach of Lucas (2010) and is estimated using the following equation:

$$\begin{aligned} Y_{iot} = \alpha + \beta_1 Post_t + \beta_2(Post_t \times DV_o^c) + \beta_3(Post_t \times Female_i) \\ + \beta_4(Post_t \times DV_o^c \times Female_i) + \text{Post}_t\, \mathbf{X}'_{i,16}\boldsymbol{\gamma} + \gamma_i + \varepsilon_{iot} \end{aligned} \tag{4}$$

where Y_{iot} is the outcome for individual i in oblast o at time t. The outcomes include four types of outcomes: labor force participation (LFP), defined as a binary variable equal to 1 if the respondent worked in the past 7 days; gender attitudes measured on a 1 – 4 Likert scale with higher values indicating stronger agreement with the statement; risk attitude; and women's decision-making indicators. The key treatment variable is baseline DV prevalence, denoted as DV_o^c, measured as the proportion of women reporting DV in each oblast in 2016 (see equations (2) and (3)). This variable serves as a proxy for the extent of exposure to the DV Law, since oblasts with higher DV prevalence before the reform are expected to be more strongly influenced by its deterrent and protective effects. The indicator $Post_t$ equals 1 in 2019 (the post-intervention period) and 0 in 2016. The variable $Female_i$ equals 1 for women and 0 for men.

The interpretation of each interaction term is as follows.

The coefficient on $Post_t$ (2019 = 1) captures the post-reform effect for men in an oblast with the average DV rate. In other words, it directly captures how outcomes for men changed after the reform in a region with average treatment intensity. The coefficient $Post_t \times DV_o^c$ shows how the post-reform effect for men varies across oblasts with different baseline DV prevalence. Specifically, it reflects how the post-reform change appears among men in oblasts with higher baseline DV prevalence. $Post_t \times Female_i$ represents the additional post-reform effect for women relative to men in oblasts with the average DV rate. In other words, it indicates how women's outcomes changed compared to men's after the reform, independent of regional DV intensity. The triple-interaction term $Post_t \times DV_o^c \times Female_i$, the key parameter of interest, captures how the post-reform effect varies with baseline DV prevalence for females relative to males, showing whether the post-reform change is more pronounced for women in oblasts with higher DV prevalence, relative to both males in the same oblasts and individuals in oblasts with lower prevalence. When the outcome is labor force participation, i.e. work variable, the coefficient β_4 represents how the law's impact on women's employment varies

with the intensity of exposure to domestic violence before the reform, that is, whether the law had a stronger effect in oblasts with higher baseline DV prevalence. These interpretations together allow the model to isolate the heterogeneous effects of the combined DV legal law by gender and by the pre-existing intensity of domestic violence across oblasts.

The motivation for this identification strategy is that, although the DV Law was implemented uniformly nationwide, I exploit regional variation in baseline DV intensity to capture differential exposure to the reform. Oblasts with higher pre-reform DV rates were more exposed to the DV Law's effects, either through stronger institutional responses or increased public awareness. This spatial variation is unlikely to be driven by short-run behavioral responses or endogenous sorting, as it primarily reflects differences in reporting infrastructure, administrative capacity, and institutional presence at the oblast level. These characteristics are slow-moving and predetermined relative to the reform, supporting the plausibility of the identification strategy.

To address unobserved heterogeneity, individual fixed effects control for time-invariant characteristics, while year fixed effects capture national-level shocks common to all regions. To improve precision, I also interact post-reform indicators with baseline household covariates (age, age square, marital status, the number of children, and consumption) measured in 2016. All these covariates are summarized in the vector $\mathbf{X}_{i,16}$ The validity of the identification relies on the parallel trends assumption: absent the reform, oblasts with different baseline DV prevalence would have followed similar trends in the outcomes. I assess this assumption using pre-trend tests with survey data from 2013 and 2016 (See A.1). Finally, given the small number of clusters (9 oblasts), inference is conducted using the Wild Cluster Bootstrap-t procedure (Cameron et al., 2008; Roodman et al., 2019), which provides more reliable p-values under limited cluster settings.

6 Results

To analyze the impact of the Domestic Violence Law, DDD analysis was conducted. Inference is based on wild cluster bootstrap with clustering at the oblast level. Standard errors are clustered at the oblast level, and p-values in parentheses are calculated using 9,999 replications. Because the data contain only nine clusters with substantial variation in cluster sizes (ranging from 266 to 1,674 observations), the conventional CRVE assumptions are violated. Therefore, I rely on wild cluster bootstrap inference (Roodman et al., 2019). In particular, I adopt the Webb weights , which perform better in small-cluster settings (fewer than 10 clusters) by providing more accurate finite-sample inference than the standard Rademacher distribution.The coefficient on Post × DV rate × Female (DDD) is the main parameter of interest, capturing the additional change experienced by women in regions with higher prereform DV prevalence, relative to men in the same regions and to both men and women in regions with lower DV prevalence. Parallel trends are shown in A.1, and it holds the assumption.

6.1 Gender attitude toward FLFP

Table 3 reports the DDD estimates for gender attitudes (measured on a disagree 1 to agree 4 scale), and the sample is restricted to those who strongly agreed, i.e., who took 4, with these statements in 2016. Columns (1) and (3) show results without baseline controls, while

Columns (2) and (4) include controls.

Across both specifications, the estimated triple-interaction terms are negative and statistically significant at the 10% level in all models, and the magnitudes remain broadly similar when controls are added, indicating robust results. In magnitude, in an oblast whose baseline DV rate is one standard deviation (0.1 percentage points) higher, women's agreement with these statements declined by about 0.10–0.16 points more than men's on the 1–4 scale. The negative and significant coefficient on (Post × Female) indicates that, on average, women became less likely than men to agree with traditional gender norms after the reform. Moreover, the negative and significant triple-interaction term suggests that females in oblasts with higher baseline DV prevalence exhibited a larger decline in agreement with traditional gender attitudes, relative to men in the same oblasts and to both men and women in low-prevalence oblasts. The coefficient on the post-reform indicator (Post) is negative and significant in the baseline models, indicating an overall decline in agreement with traditional gender views between 2016 and 2019. However, once individual and household characteristics are interacted with the post indicator, the coefficient becomes statistically insignificant, suggesting that much of this aggregate change can be explained by compositional shifts in demographic and socioeconomic attributes rather than by the reform itself.

Overall, restricting the sample to individuals who strongly endorsed traditional gender norms in 2016, women in oblasts with higher pre-reform DV prevalence exhibited the largest attitudinal improvements after the reform, that is they were less likely to agree that women should not work outside the home due to religious considerations (No_WWOH_reli) and that the husband's career should be prioritized (Huscareer). Furthermore, women overall exhibited significantly larger attitudinal shifts away from traditional gender norms than men, indicating a broad-based improvement in gender attitudes following the reform.

Two caveats apply. First, treating the two attitude outcomes as one family, the DDD estimates remain significant at the 10% level after the Benjamini–Hochberg correction ($q = 0.095$) but not after the more conservative Holm correction ($p = 0.119$; Appendix Table 15). Second, when the sample is not restricted to respondents who strongly agreed in 2016, the DDD estimates are negative but small and insignificant (Appendix Table 10). The attitudinal response is thus concentrated among initially conservative respondents rather than spread across the population. Because this sample starts at the top of the 1–4 scale, its outcomes can only decline, so the large negative Post coefficients partly reflect mean reversion; the DDD estimates are informative to the extent that such reversion does not differ by gender and baseline DV prevalence.

Table 3: DDD Results for Gender Attitudes Toward Female Labor Force Participation

	Women should not work outside home		Husband's career comes first	
	(1) No controls	(2) Controls	(3) No controls	(4) Controls
Post (2019 = 1)	-1.052***	-1.017	-1.044***	-0.124
	(0.003)	(0.574)	(0.000)	(0.912)
Post × DV rate	-0.172	-0.169**	-0.076	-0.068
	(0.120)	(0.048)	(0.348)	(0.512)
Post × Female	-0.256***	-0.273***	-0.164***	-0.169***
	(0.004)	(0.003)	(0.003)	(0.004)
Post × DV rate × Female (DDD)	-0.118*	-0.096*	-0.160*	-0.157*
	(0.095)	(0.074)	(0.060)	(0.081)
Num children (baseline) × Pre		-0.032***		-0.007
		(0.008)		(0.586)
Age (baseline) × Pre		0.016**		0.007
		(0.017)		(0.354)
Age^2 (baseline) × Pre		-0.000		-0.000
		(0.133)		(0.394)
Log food cons. (baseline) × Pre		0.017		0.022
		(0.912)		(0.825)
Log non-food cons. (baseline) × Pre		-0.041		0.043
		(0.562)		(0.306)
Married (baseline) × Pre		0.020		-0.041
		(0.709)		(0.638)
Individual FE	Yes	Yes	Yes	Yes
Observations	1346	1306	1686	1616
Clusters	9	9	9	9
Replications	9999	9999	9999	9999

Notes: Columns (1) and (3) report DDD estimates without baseline controls; columns (2) and (4) include interactions of the post-reform indicator with baseline household covariates: age, age squared, marital status, number of children, food consumption, and non-food consumption. The outcome in columns (1) and (2) is an indicator for agreeing that a woman should not work outside the home due to religious considerations. The outcome in columns (3) and (4) is an indicator for agreeing that the husband's career should be prioritized over the wife's career. All regressions absorb individual fixed effects and include a post-reform indicator. Baseline covariates are interacted with the pre-reform (2016) indicator; with individual fixed effects, the post-reform interaction equals minus the reported coefficient. The DV rate is scaled so that one unit equals 0.1 percentage points. Standard errors are clustered at the oblast level, with 9 clusters. Reported p-values are wild cluster bootstrap p-values with 9,999 replications. Stars denote significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Risk attitude and Females' decision making

Table 4 reports the DDD estimates for risk attitude and female decision-making.

For the risk attitude, the Post term is –0.472, significant at the 1% level, indicating that risk attitude decreased by nearly half of a standard deviation between 2016 and 2019 for the men in the average oblast. At the same time, the DDD coefficient (Post × DV rate × Female) is +0.030 and significant at the 10% level. This positive coefficient reflects a smaller decline among women than among men in higher-DV-prevalence regions, rather than an increase in women's willingness to take risks: the gender-specific estimate for women is negative and insignificant (Appendix Table 11). After controlling for baseline household characteristics, the coefficient on Post becomes positive but statistically insignificant. This suggests that the earlier negative effect on risk attitude may have been influenced by baseline differences rather than the reform itself.

The decision-making index is constructed as the share of household decisions (out of 17) in which at least one female household member participated Female discision making index takes 1, and serves as a proxy for women's bargaining power.The results indicate that women significantly improved their decision-making power after the reform relative to men (Post × Female), and this effect remains positive and significant when controls are added. Furthermore, the positive and statistically significant DDD coefficient in Column (4) suggests that women in higher-DV-prevalence regions experienced particularly large gains in decision-making power relative to women in low-prevalence regions and men.

Overall, these results provide inconclusive evidence regarding changes in risk attitude but robust and consistent improvements in women's intra-household decision-making power following the reform.

Because the DDD coefficient measures the change for women *relative to men*, it does not by itself show whether women's own outcomes changed. Appendix Table 11 therefore reports gender-specific effects of the DV rate. For the statement that the husband's career should come first, the effect for women is negative and significant at the 5% level with and without controls (-0.236, $p = 0.031$; -0.225, $p = 0.020$), indicating that women in higher-DV oblasts themselves moved away from this norm. Women's decision-making index also rose in higher-DV oblasts (0.040, $p = 0.080$; 0.049, $p = 0.070$). For the statement that women should not work outside the home, the effect for women is significant only with controls (-0.265, $p = 0.085$). For risk attitudes and work participation, the effects for women are insignificant.

Table 4: DDD Results for Risk Attitudes and Female Decision-Making

	Risk attitude index		Female decision-making index	
	(1) No controls	(2) Controls	(3) No controls	(4) Controls
Post (2019 = 1)	-0.472***	0.752	-0.019	1.088**
	(0.003)	(0.832)	(0.876)	(0.022)
Post × DV rate	-0.292	-0.247	0.023	0.030
	(0.250)	(0.343)	(0.183)	(0.159)
Post × Female	0.055	0.037	0.043**	0.039*
	(0.210)	(0.399)	(0.029)	(0.056)
Post × DV rate × Female (DDD)	0.030*	0.020	0.017	0.019**
	(0.086)	(0.442)	(0.161)	(0.027)
Num children (baseline) × Pre		-0.050		-0.014**
		(0.268)		(0.028)
Age (baseline) × Pre		0.019		0.004
		(0.230)		(0.318)
Age^2 (baseline) × Pre		-0.000		-0.000
		(0.153)		(0.238)
Log food cons. (baseline) × Pre		0.262		0.059*
		(0.374)		(0.083)
Log non-food cons. (baseline) × Pre		-0.150		0.026
		(0.481)		(0.214)
Married (baseline) × Pre		-0.052		0.034
		(0.464)		(0.312)
Individual FE	Yes	Yes	Yes	Yes
Observations	6874	6734	6874	6734
Clusters	9	9	9	9
Replications	9999	9999	9999	9999

Notes: Columns (1) and (3) report DDD estimates without baseline controls; columns (2) and (4) include interactions of the post-reform indicator with baseline household covariates: age, age squared, marital status, number of children, food consumption, and non-food consumption. The risk attitude index is standardized using the 2016 baseline distribution. All regressions absorb individual fixed effects and include a post-reform indicator. Baseline covariates are interacted with the pre-reform (2016) indicator; with individual fixed effects, the post-reform interaction equals minus the reported coefficient. The DV rate is scaled so that one unit equals 0.1 percentage points. Standard errors are clustered at the oblast level, with 9 clusters. Reported p-values are wild cluster bootstrap p-values with 9,999 replications. Stars denote significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.3 FLFP

Table 5 reports DDD estimates for work participation in the past seven days. The triple interaction term (Post × DV rate × Female) is not statistically significant, implying that women in higher-DV-prevalence regions did not experience a differential increase compared to women in lower-prevalence regions. However, female labor participation increased slightly but significantly relative to men after the reform (Post × Female). This is still robust at the 5% level when adding the control terms. Although male labor participation (Post term) did not show a statistically significant change, the coefficient suggests a modest declining trend.

Taken together, the evidence suggests that female labor force participation increased significantly after the reform, as women became more likely to work relative to men. However, this overall increase cannot be clearly attributed to the DV Law, since the DDD coefficient is insignificant, and other macro-level factors such as economic conditions or family-support policies may also have contributed to the observed rise. In particular, the law did not generate measurable heterogeneous effects across regions with different levels of baseline DV prevalence.

While the aggregate triple-difference effect on *Worked in past 7 days* is statistically insignificant, this aggregate may conceal offsetting movements across types of work and across different groups of women. As defined above, *Worked in past 7 days* is a broad indicator, so by construction it cannot reveal which type of work responded to the reform, nor which women drove the response. I therefore probe beneath the aggregate in the next subsection.

Table 5: DDD Results for Work: With and Without Controls

	Worked in past 7 days	
Post (2019 = 1)	0.024	0.305
	(0.702)	(0.637)
Post × DV rate	-0.091	-0.096
	(0.304)	(0.344)
Post × Female	0.036*	0.039**
	(0.070)	(0.040)
Post × DV rate × Female (DDD)	0.051	0.057
	(0.160)	(0.160)
Num children (baseline) × Pre		-0.004
		(0.463)
Age (baseline) × Pre		0.008*
		(0.078)
Age^2 (baseline) × Pre		-0.000
		(0.170)
Log food cons. (baseline) × Pre		0.003
		(0.946)
Log non-food cons. (baseline) × Pre		0.005
		(0.823)
Married (baseline) × Pre		0.012
		(0.624)
Individual FE	Yes	Yes
Observation	6874	6734
clusters	9	9
reps	9999	9999

Notes: Columns (1) and (2) report DDD estimates for work participation in the past 7 days. Column (1) excludes baseline controls, while Column (2) includes interactions of the post-reform indicator with baseline household covariates (age, age squared, marital status, number of children, and consumption). All regressions absorb individual fixed effects and include a post-reform indicator. Baseline covariates are interacted with the pre-reform (2016) indicator; with individual fixed effects, the post-reform interaction equals minus the reported coefficient. The DV rate is scaled so that one unit equals 0.1 percentage points. Standard errors are clustered at the oblast level. Reported p-values are wild cluster bootstrap p-values with 9,999 replications. Stars denote significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.4 Heterogeneity in Employment Outcomes

These employment-history outcomes complement work7 with a longer view: from the Life in Kyrgyzstan calendar I construct, for each respondent, the number of months in the past twelve spent in agricultural and other work, months as a housewife, and indicators for the main activity over the year (Appendix Table 12). Because they summarize the dominant activity across a year rather than any activity in a single week, these measures are complementary to

the seven-day indicator, not a decomposition of it. I estimate the DDD for these outcomes on the full sample and on subsamples (married, below 40, and urban at baseline); the significant estimates appear in Tables 6 and 7, and the full set in Appendix Table 13.

In the full sample (Table 6), the response is concentrated in other work. The DDD estimate for other-work months is positive and marginally significant: a 0.1-percentage-point higher baseline DV rate is associated with an additional 0.53 months of other work for women relative to men after the reform. By contrast, agricultural months, housewife months, and total employment months show no comparable response. The main-activity indicator for other work is also positive, although not statistically significant. This suggests that the reform did not broadly increase women's employment, but may have shifted the composition of work toward other activities in higher-DV-prevalence regions.

This pattern is stronger among married women (Table 7). In the married subsample, the seven-day employment indicator becomes marginally significant, implying a 6.1–6.5 percentage-point increase for women relative to men in regions with a 0.1-percentage-point higher baseline DV rate. The effect on other-work months is also larger, at about 0.55–0.61 additional months, and the indicator for other work as the main annual activity increases by about 4.8–5.3 percentage points. These estimates are stable with and without baseline controls. This coherence is notable given the law's design. The 2017 Family Violence Law extends protection to family members and persons equated to them (Baker McKenzie, nd); its definition of "family" has been criticized for excluding unmarried partners (Watch, 2019). Its safeguards therefore apply most directly to women in marital and family relationships—precisely the group in which the employment response concentrates. That the effect surfaces where the law can actually bite is reassuring for a bargaining interpretation, even though the interaction tests indicate that the gap between married and unmarried women is not itself sharply estimated. By contrast, the young and urban subsamples do not show a significant aggregate employment response. These results are consistent with the bargaining interpretation, since married women are the group for whom changes in legal protection within the household are most directly relevant.

An interaction test sharpens where this shift sits. Adding to the full sample a quadruple interaction with each baseline characteristic in turn (urban residence, marital status, age), only the urban term—Post $\times$ Female $\times$ DV rate $\times$ Urban—on *Other work as main activity* is marginally significant (0.026, $p = 0.068$; Table 8); the marital-status and age terms are not (full set in Appendix Table 14). The reallocation is thus concentrated among women who are both in higher-DV-prevalence oblasts—those most exposed to the reform under the intensity design—and in urban areas. The labor response emerges at the intersection of high treatment intensity and urban residence, rather than reflecting general urban progressiveness.

Taken together, these results sharpen rather than overturn the aggregate null. The reform's labor-market footprint is not a broad rise in employment but a narrower reallocation—concentrated among married women in higher-DV regions—toward other work. This is the margin one would expect if the mechanism runs through women's intra-household bargaining position and autonomy rather than through aggregate labor demand, since other work is less home-tethered and more income- and autonomy-conferring than the subsistence and unpaid family labor that dominate the seven-day aggregate. That aggregate, pooling all activity types into a single snapshot, can therefore look flat even as the annual composition of women's work shifts toward other work. I read these patterns as suggestive rather than confirmatory: as Appendix Table 15 shows, the other work estimates do not survive multiple-hypothesis correction, so they are best interpreted as descriptive evidence on the channel of adjustment behind

the aggregate result.

Table 6: Selected DDD Estimates for Employment Outcomes: Full Sample

	Other-work months	Other work as main activity
No controls	0.527*	0.044
	(0.062)	(0.116)
With baseline controls	0.555	0.046
	(0.113)	(0.175)
Individual FE	Yes	Yes
Clusters	9	9

Notes: Other-work months count months in the past 12 months spent in non-agricultural or other work activities. Other work as main activity equals one if other work accounts for strictly more observed months than agricultural work and housewife status; ties are coded as zero. Wild bootstrap p-values are reported in parentheses. Wild bootstrap uses Webb weights with 9,999 replications. Stars are based on wild bootstrap p-values. Standard errors are clustered at the oblast level, with 9 clusters. DV rate is scaled so that one unit equals 0.1 percentage points. Reported coefficients are for Post × Female × DV rate. Baseline controls include age, age squared, marital status, number of children, and household food and non-food consumption, each interacted with the post-period indicator. The full set of outcomes is reported in Appendix Table 13. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Selected DDD Estimates for Employment Outcomes: Married Subsample

	Worked in past 7 days	Other-work months	Other work as main activity
No controls	0.061*	0.551**	0.048*
	(0.092)	(0.047)	(0.086)
With baseline controls	0.065*	0.605*	0.053*
	(0.093)	(0.057)	(0.097)
Individual FE	Yes	Yes	Yes
Clusters	9	9	9

Notes: Other-work months count months in the past 12 months spent in non-agricultural or other work activities. Other work as main activity equals one if other work accounts for strictly more observed months than agricultural work and housewife status; ties are coded as zero. Wild bootstrap p-values are reported in parentheses. Wild bootstrap uses Webb weights with 9,999 replications. Stars are based on wild bootstrap p-values. Standard errors are clustered at the oblast level, with 9 clusters. DV rate is scaled so that one unit equals 0.1 percentage points. Reported coefficients are for Post × Female × DV rate. Baseline controls include age, age squared, marital status, number of children, and household food and non-food consumption, each interacted with the post-period indicator. The married subsample is restricted to individuals married at baseline. The full set of outcomes is reported in Appendix Table 13. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Selected Heterogeneity Interaction Test

	Other work as main activity
Urban at baseline	0.026*
	(0.068)
Baseline controls	No
Individual FE	Yes
Clusters	9

Notes: Wild bootstrap p-values are reported in parentheses. Wild bootstrap uses Webb weights with 9,999 replications. Stars are based on wild bootstrap p-values. Individual fixed effects are absorbed. Standard errors are clustered at the oblast level, with 9 clusters. DV rate is scaled so that one unit equals 0.1 percentage points. The reported coefficient is the coefficient on the quadruple interaction between Post, Female, DV rate, and urban residence. The full set of interaction tests is reported in Appendix Table 14. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.5 Results Summary

Among individuals who were initially highly conservative, women, especially those in oblasts with higher pre-reform DV prevalence, showed the largest improvements in gender attitudes toward FLFP after the Domestic Violence Law. In addition, women overall became less likely to support restrictive views on female labor, indicating a broader attitudinal shift beyond regional differences in DV prevalence.

By contrast, female labor force participation rose relative to men after the reform, indicating a general improvement in women's employment outcomes. However, this change cannot be directly attributed to the DV Law, as no significant heterogeneity was found across regions with different levels of baseline DV prevalence. Disaggregating this aggregate, however, reveals movement beneath the null: the composition of women's work shifts toward other work in higher-DV regions, and this reallocation is concentrated among married women—the group most directly covered by a law that protects family relationships. These employment patterns do not survive multiple-hypothesis correction and are therefore suggestive rather than confirmatory, but they are consistent with the same bargaining channel that operates on attitudes and decision-making: a participation measure that pools subsistence and unpaid family labor can look flat even as women reallocate toward more autonomy-conferring work.

For risk attitude, results are inconclusive. Although women in high-DV regions appeared relatively less likely to reduce their willingness to take risks, the effect is not robust once baseline controls are included. In contrast, household decision-making shows a clearer pattern: women gained greater bargaining power compared to men.

These findings are largely consistent with the theoretical framework, which posits that the DV Law influences women's labor decisions through two intermediate mechanisms: enhancing intra-household bargaining power and shifting social norms toward female employment. Both mechanisms are supported by the results.However, the aggregate labor response remains muted: any movement is confined to a compositional shift toward other work among married

women, rather than a broad, DV-attributable rise in participation.

7 Discussion and Limitation

7.1 Limitation

A limitation of this design is that I cannot separately identify the effects of the 2017 and 2019 reforms. The 2017 reform strengthened victim protection, while the 2019 reform increased the legal costs for perpetrators by criminalizing domestic violence. These two components may have operated through different channels. Victim protection may have improved women's bargaining power, whereas criminalization may have generated backlash or heightened intra-household tensions. Therefore, the estimates should be interpreted as the net effect of the combined reforms rather than the separate effect of each legal change.
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The empirical strategy relies on regional variation in pre-reform domestic violence prevalence to identify the impact of the legal reforms, suggesting that this variation may not be fully exogenous. In particular, pre-reform DV intensity may reflect deeper, long-standing differences in social norms or institutional quality across regions, which cannot be fully disentangled in this setting. Although year fixed effects absorb nationwide time trends, the estimates may still capture the influence of other concurrent policies or broader social developments that occurred during the same period. As such, the reported effects may be interpreted as an upper bound of the impact of the DV Law, admitting potential upward bias. The pure effect of the reform is therefore likely to be smaller than the estimated coefficients. Accordingly, the estimated coefficients should be interpreted as relative regional effects rather than precise nationwide causal effects.

Second point is data availability and measurement constraints. Migration and remittances play a significant role in household livelihoods in Kyrgyzstan; however, information on remittance receipt is available for only about 13% of the sample. Moreover, among respondents who report this information, more than 90% indicate receiving remittances, suggesting substantial selective reporting. Due to this limitation, remittance income cannot be reliably incorporated into the empirical analysis.

Third, the dataset does not capture relationship arrangements beyond formal marriage within the household, such as cohabitation, informal partnerships, or past marital histories. These unobserved relationship statuses may affect exposure to domestic violence and legal protection but cannot be accounted for in the present analysis, introducing unobserved heterogeneity.

Finally, structural characteristics of local labor markets are not fully captured in the data. During the survey period, Kyrgyzstan still maintained a legal list of occupations from which women were excluded; this list was abolished only in 2025. The restrictions, however, applied mainly to physically demanding or hazardous jobs such as underground mining, and most occupations were open to women. Formal occupational segregation is therefore unlikely to have been a major constraint on the employment margins studied here, although it remained in place. Other features of the labor market, such as the prevalence of informal employment and employer bias, may constrain women's labor market responses to changing norms and

legal protections more directly. These structural constraints may partially limit the extent to which normative changes translate into observed employment outcomes.

7.2 Future research

This study points to several promising directions for future research. In particular, further work could distinguish more clearly between public gender norms and private, spouse-specific norms within households. While the present analysis captures public attitudes and indicators of intra-household decision-making, richer data would allow researchers to examine how beliefs about one's own spouse may differ from publicly expressed norms.

Future research using more detailed attitudinal surveys, paired spouse interviews, or experimental designs could investigate whether legal reforms affect public and private norms differently, and how such divergences shape women's labor supply and household bargaining dynamics. Understanding these mechanisms would provide deeper insight into the channels through which legal protection translates—or fails to translate—into economic behavior.

8 Conclusion

This study examined the effects of the combined domestic violence (DV) legal reforms in Kyrgyzstan—including the 2017 Law on Prevention and Protection Against Domestic Violence and the subsequent 2019 criminalization—on gender norms and female labor force participation (FLFP), using an intensity-based difference-in-difference-in-differences framework. By combining attitudinal indicators with behavioral outcomes, the analysis explored how strengthened legal protections interacted with persistent traditional norms and uneven institutional capacity.

The findings suggest that the DV legal reforms were associated with meaningful changes in gender attitudes and intra-household dynamics. Among individuals with initially conservative views, women—particularly those residing in oblasts with higher pre-reform DV prevalence—became less likely to endorse restrictive norms regarding female labor. Women in these regions also experienced increased involvement in household decision-making, consistent with an improvement in bargaining positions within the household. Changes in risk attitudes point in a similar direction, although these effects are weaker and sensitive to the inclusion of baseline controls.

In contrast, behavioral responses in the labor market appear more limited. While female labor force participation increased relative to men in the post-reform period, this change does not exhibit clear heterogeneity across regions with different levels of pre-reform DV prevalence. A closer look nevertheless reveals a narrower response: in higher-DV regions, married women reallocate toward other work even as aggregate participation remains flat. This compositional shift is suggestive rather than conclusive, but it is the margin one would expect if the reform operated through women's intra-household bargaining position rather than through aggregate labor demand. This pattern suggests that the impact of the legal reforms was concentrated primarily in normative and intra-household domains, rather than translating directly into labor market participation.

Taken together, these results indicate that strengthened domestic violence legislation can influence women's attitudes and bargaining environments even when immediate labor market

responses remain muted. The findings highlight the importance of institutional complementarity: legal reforms alone may be insufficient to generate sustained economic empowerment unless accompanied by complementary policies, such as accessible childcare, equal employment opportunities, and effective enforcement of anti-discrimination protections. In contexts like Kyrgyzstan, where social norms and labor market constraints remain binding, translating perceptual change into behavioral and economic outcomes requires a broader policy ecosystem.

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A Appendix

A.1 Pre-trend test

This section presents the pre-trend analysis. Since merging the three survey years (2013, 2016, and 2019) leads to substantial missing values, I employed a placebo test using a two-way fixed effects DDD approach between the 2013 and 2016 waves, with the placebo treatment assigned to the year 2015 (Table 9). The DDD estimator removes common biases by differencing two DID estimators. Therefore, its identification requires only one assumption: the relative parallel trends assumption, namely that in the absence of treatment, the gender gap would have evolved similarly across treatment and control regions (see Gruber 1994 for an early application; Olden and Møen 2022 for a formal discussion; Ortiz-Villavicencio and Sant'Anna 2025 for recent extensions).

Across all outcome variables, none of the DDD interaction terms are statistically significant at conventional levels although some lower-order terms are significance. Since the pre-trend test specifically concerns the DDD interactions, this indicates that prior to the enactment of the 2017 Domestic Violence Law, there were no differential trends between treatment and control groups. These findings support the unconditional parallel trends assumption. A limitation is that the test relies only on 2013 and 2016, so longer-term pre-trends cannot be fully assessed.

Table 9: DDD Results of Pre-Trend Placebo Test

	(1) No_WWOH_reli	(2) Huscareer	(3) Risk std	(4) Fem DM	(5) Work index
Post (=1)	-1.324**	-1.027***	-0.006	0.008**	0.081*
	(0.012)	(0.004)	(0.956)	(0.011)	(0.084)
Post × DV rate	1.067	0.806	-0.468	0.001	-0.284
	(0.318)	(0.625)	(0.701)	(0.917)	(0.214)
Post × Female	-0.123	-0.228*	-0.019	0.015***	-0.030
	(0.334)	(0.050)	(0.832)	(0.000)	(0.169)
Post × DV rate × Female	-0.612	-0.516	0.493	0.005	0.107
	(0.111)	(0.245)	(0.182)	(0.570)	(0.237)
Obs	2176	2786	7168	7636	7168
clusters	9	9	9	9	9
reps	9999	9999	9999	9999	9999

Notes: Each column reports results from a DDD specification between 2013 and 2016 waves for the pre-trend test. The outcomes are: (1) belief that woman should not work outside home due to religious considerations, (2) belief that husband's career should be more prioritized than wife, (3) risk attitude standardized with 2016 as the baseline, (4) female decision-making index and (5) any work participation past 7 days. All regressions absorb individual fixed effects and include a post-reform indicator; standard errors are clustered at the oblast level. The 2013 DV rate is centered and scaled so that one unit equals 0.1 percentage points. Reported p-values are wild cluster bootstrap p-values with 9,999 replications. Stars denote significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2 Gender attitudes in the full sample

Table 10: DDD Results for Gender Attitudes: Full Sample

	Women should not work outside home		Husband's career comes first	
	(1) No controls	(2) Controls	(3) No controls	(4) Controls
Post (2019 = 1)	0.333***	0.060	0.108	-1.206
	(0.002)	(0.974)	(0.552)	(0.418)
Post × DV rate	-0.046	-0.067	-0.099	-0.117
	(0.773)	(0.718)	(0.360)	(0.314)
Post × Female	0.006	-0.003	0.046	0.031
	(0.915)	(0.968)	(0.537)	(0.670)
Post × DV rate × Female (DDD)	-0.028	-0.033	-0.036	-0.045
	(0.339)	(0.266)	(0.304)	(0.239)
Baseline controls × Pre	No	Yes	No	Yes
Individual FE	Yes	Yes	Yes	Yes
Observations	6874	6734	6874	6734
Clusters	9	9	9	9
Replications	9999	9999	9999	9999

Notes: Same specifications as Table 3, estimated on the full analysis sample instead of respondents who strongly agreed with each statement in 2016. Baseline controls are as in Table 3. All regressions absorb individual fixed effects; standard errors are clustered at the oblast level. Wild cluster bootstrap p-values with Webb weights and 9,999 replications are reported in parentheses. Stars denote significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.3 Gender-specific effects

Table 11: Gender-Specific Effects of the DV Rate

	No WWOH		Husband career		Risk		Female DM		Work	
Controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Post × DV rate (men)	-0.172	-0.169**	-0.076	-0.068	-0.292	-0.247	0.023	0.030	-0.091	-0.096
	(0.120)	(0.048)	(0.348)	(0.512)	(0.250)	(0.343)	(0.183)	(0.159)	(0.304)	(0.344)
Post × DV rate (women)	-0.289	-0.265*	-0.236**	-0.225**	-0.262	-0.227	0.040*	0.049*	-0.040	-0.039
	(0.137)	(0.085)	(0.031)	(0.020)	(0.279)	(0.334)	(0.080)	(0.070)	(0.443)	(0.508)
Post (men)	-1.052***	-1.017	-1.044***	-0.124	-0.472***	0.752	-0.019	1.088**	0.024	0.305
	(0.003)	(0.574)	(0.000)	(0.912)	(0.003)	(0.832)	(0.876)	(0.022)	(0.702)	(0.637)
Post (women)	-1.308***	-1.289	-1.208***	-0.294	-0.417***	0.789	0.024	1.127**	0.060**	0.345
	(0.002)	(0.473)	(0.000)	(0.799)	(0.004)	(0.821)	(0.569)	(0.018)	(0.046)	(0.607)
DDD (women − men)	-0.118	-0.096	-0.160	-0.157	0.030	0.020	0.017	0.019	0.051	0.057
wild bootstrap p-value	0.095	0.074	0.060	0.081	0.086	0.442	0.161	0.027	0.160	0.160
Observations	1346	1306	1686	1616	6874	6734	6874	6734	6874	6734
Clusters	9	9	9	9	9	9	9	9	9	9

Notes: This table reports gender-specific estimates from a reparametrized version of the DDD models in Tables 3–5, with separate Post and Post × DV rate terms for men and women. Post × DV rate (women) equals the sum of Post × DV rate and Post × DV rate × Female in the DDD model, and the DDD row reports the difference between women and men, which reproduces the triple-interaction coefficients in Tables 3–5. The DV rate is mean-centered and scaled so that one unit equals 0.1 percentage points. Samples and controls are as in Tables 3–5. All regressions absorb individual fixed effects; standard errors are clustered at the oblast level. Wild cluster bootstrap p-values with Webb weights and 9,999 replications are reported in parentheses. Stars denote significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4 Outcome definitions

Table 12: Definition of Employment-History Outcomes

Outcome	Definition
Employment months	Number of months in any work activity[a].
Agricultural work months	Number of months in agricultural work[b].
Other-work months	Number of months in other work[c].
Housewife months	Number of months in which the respondent's main activity is housewife.
Main agricultural work	Indicator equal to one if agricultural work accounts for more months than both other work and housewife status.
Main other work	Indicator equal to one if other work accounts for more months than both agricultural work and housewife status.
Main housewife	Indicator equal to one if housewife status accounts for more months than both agricultural work and other work.

Notes: Monthly activities are based on respondents' reported main activity in each of the past 12 months. Main activity indicators require the corresponding category to account for strictly more observed months than the other two categories; ties are coded as zero.
[a] Work activity includes agricultural work and other work.
[b] Agricultural work includes agricultural wage work and agricultural own-account work.
[c] Other work includes non-agricultural wage work, non-agricultural own-account work, employers, members of producers' cooperatives, and contributing family workers.

A.5 Heterogeneity check

A.5.1 Subsample Analysis

Table 13: DDD Estimates for Employment Outcomes across Subsamples

	(1) work7	(2) emp_months	(3) agri_months	(4) other_work	(5) housewife	(6) main_agri	(7) main_other	(8) main_housewife
Panel A. Full sample								
No controls	0.052	0.411	-0.116	0.527*	0.274	-0.016	0.044	0.016
	(0.170)	(0.858)	(0.737)	(0.062)	(0.450)	(0.607)	(0.116)	(0.412)
With baseline controls	0.056	0.436	-0.119	0.555	0.236	-0.016	0.046	0.012
	(0.178)	(0.848)	(0.729)	(0.113)	(0.493)	(0.592)	(0.175)	(0.430)
Panel B. Married subsample								
No controls	0.061*	0.539	-0.013	0.551**	0.283	-0.011	0.048*	0.014
	(0.092)	(0.580)	(0.962)	(0.047)	(0.502)	(0.671)	(0.086)	(0.411)
With baseline controls	0.065*	0.560	-0.045	0.605*	0.267	-0.014	0.053*	0.012
	(0.093)	(0.615)	(0.862)	(0.057)	(0.506)	(0.617)	(0.097)	(0.476)
Panel C. Young subsample								
No controls	0.046	0.642	0.138	0.504	0.463	0.006	0.030	0.026
	(0.371)	(0.577)	(0.793)	(0.364)	(0.555)	(0.889)	(0.590)	(0.453)
With baseline controls	0.055	0.730	0.080	0.650	0.407	0.001	0.043	0.020
	(0.155)	(0.602)	(0.836)	(0.340)	(0.594)	(0.965)	(0.416)	(0.584)
Panel D. Urban subsample								
No controls	0.051	0.270	-0.291	0.561	-0.410	-0.034	0.047	-0.026
	(0.408)	(0.264)	(0.343)	(0.136)	(0.312)	(0.335)	(0.110)	(0.322)
With baseline controls	0.050	0.201	-0.307	0.508	-0.442	-0.035	0.044	-0.031
	(0.418)	(0.277)	(0.327)	(0.138)	(0.305)	(0.325)	(0.115)	(0.303)

Notes: Wild bootstrap p-values are reported in parentheses. Wild bootstrap uses Webb weights with 9,999 replications. Stars are based on wild bootstrap p-values. Individual fixed effects are absorbed. All specifications cluster standard errors at the oblast level, with 9 clusters. DV rate is scaled so that one unit equals 0.1 percentage points. Baseline controls include age, age squared, marital status, number of children, and household food and non-food consumption, each interacted with the post-period indicator. The married subsample is restricted to individuals married at baseline; the young subsample to individuals below age 40 at baseline; and the urban subsample to individuals living in urban areas at baseline. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.5.2 Interaction Test

Table 14: Heterogeneity Interaction Tests for Employment Outcomes

	(1) work7	(2) emp_months	(3) agri_months	(4) other_work	(5) housewife	(6) main_agri	(7) main_other	(8) main_housewife
Married at baseline	0.028	0.279	0.121	0.158	0.227	-0.011	0.031	0.012
	(0.616)	(0.728)	(0.715)	(0.630)	(0.440)	(0.785)	(0.363)	(0.509)
Below age 40 at baseline	-0.017	0.354	0.452	-0.099	0.351	0.038	-0.029	0.020
	(0.581)	(0.558)	(0.206)	(0.927)	(0.589)	(0.126)	(0.872)	(0.630)
Urban at baseline	-0.008	-0.283	-0.570	0.287	-0.864	-0.053	0.026*	-0.049
	(0.792)	(0.819)	(0.784)	(0.132)	(0.132)	(0.830)	(0.068)	(0.258)
Baseline controls	No	No	No	No	No	No	No	No
Clusters	9	9	9	9	9	9	9	9

Notes: All specifications are estimated without baseline controls. Wild bootstrap p-values are reported in parentheses. Wild bootstrap uses Webb weights with 9,999 replications. Stars are based on wild bootstrap p-values. Individual fixed effects are absorbed. Standard errors are clustered at the oblast level, with 9 clusters. DV rate is scaled so that one unit equals 0.1 percentage points. Each row reports the coefficient on the quadruple interaction between Post, Female, DV rate, and the listed baseline characteristic. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.5.3 Multiple Hypothesis Testing Adjustments

Because multiple related outcomes are tested, I also report Holm-adjusted p-values and Benjamini–Hochberg false discovery rate q-values by outcome family in Table 15.

Table 15: Multiple Hypothesis Testing Adjustments by Outcome Family

Outcome	Wild bootstrap p	Holm adjusted p	BH-FDR q
Panel A. Overall employment outcomes			
work7	0.170	0.339	0.339
emp_months	0.858	0.858	0.858
Panel B. Monthly activity outcomes			
other_work_months	0.062	0.186	0.186
housewife_months	0.450	0.899	0.674
agri_months	0.737	0.899	0.737
Panel C. Main activity dummy outcomes			
main_other_work_12m	0.116	0.348	0.348
main_housewife_12m	0.412	0.823	0.607
main_agri_12m	0.607	0.823	0.607
Panel D. Gender norm outcomes			
No_WWOH_reli	0.095	0.119	0.095
Huscareer_imp_wife	0.060	0.119	0.095

Notes: This table reports multiple hypothesis testing adjustments by outcome family. The Wild bootstrap p-values are the original, unadjusted p-values for the DDD coefficient in each outcome-specific regression. Holm adjusted p-values control the family-wise error rate, and Benjamini–Hochberg false discovery rate q-values control the expected false discovery rate within each outcome family. Outcome families are defined as overall employment outcomes, monthly activity outcomes, main activity dummy outcomes, and gender norm outcomes. Panels A–C use the full-sample specifications without baseline controls in Panel A of Appendix Table 13; Panel D uses the specifications without baseline controls in columns (1) and (3) of Table 3, restricted to respondents who strongly agreed with the statement in 2016. Wild bootstrap p-values use Webb weights with 9,999 replications.